

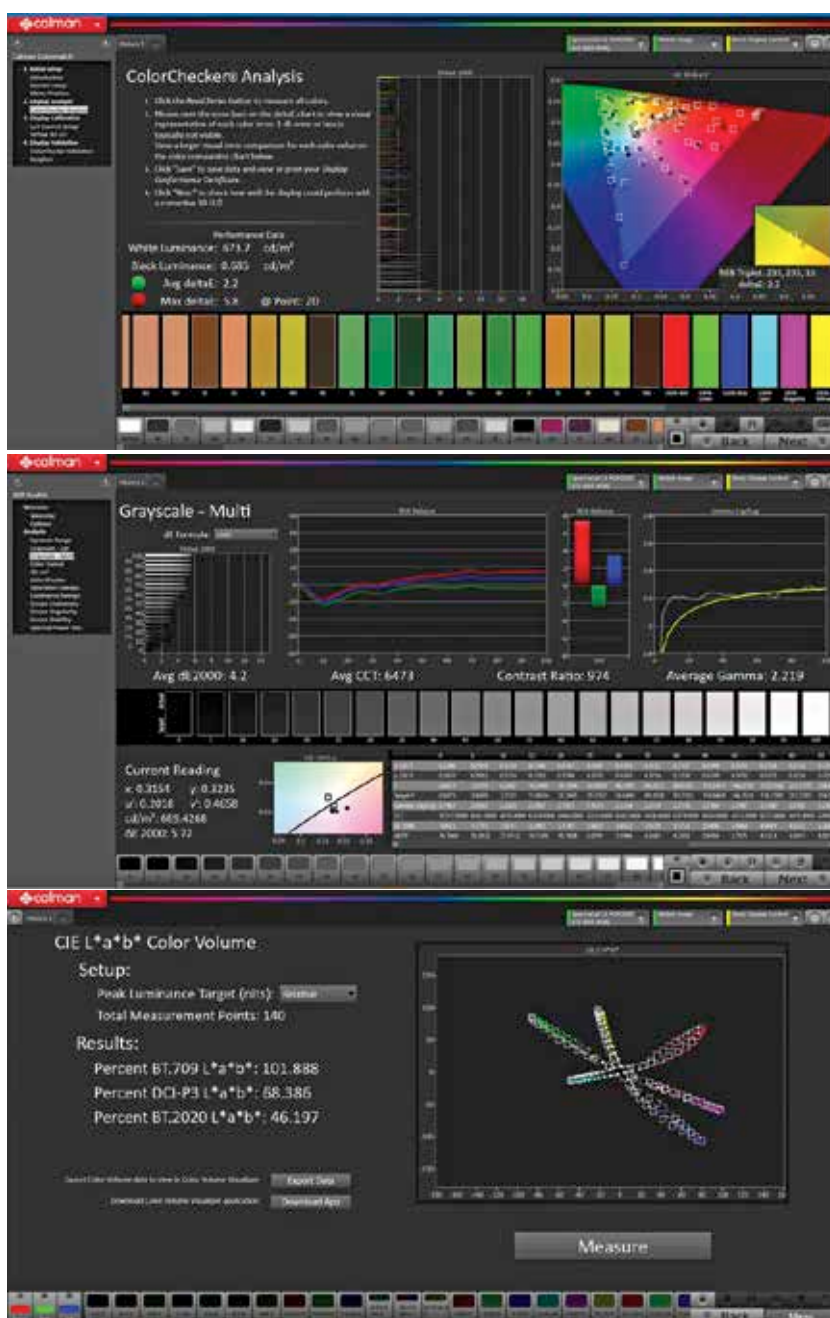
Elon Musk launches 'Optimus'

Elon Musk unveiled a prototype of a humanoid "Optimus" robot that shares some AI software.
<https://digit.in/oct22-05>



OpenAI removes DALL-E waitlist

Users can sign up and start using DALL-E immediately without any waiting period.
<https://digit.in/oct22-06>



that's the basic principle of how an FFS display differs from an IPS LCD display.

Almost all displays in use today are active matrix displays that require a capacitor for every pixel to hold charge till the next refresh cycle. Another advantage of FFS mode is that this storage capacitor (Cst as shown in the image above) is automatically formed in each pixel owing to overlap between the pixel and the common electrodes as shown in the image above. All other LCD modes need to dedicate

some pixel space for the capacitor, resulting in lesser transmittance. This gives FFS an advantage when high-resolution screens with small pixel-size are needed.

FFS LCD DISPLAY PERFORMANCE IN PRACTICE

Well, now that we have discussed the theory, let's talk about the practical experience. We tested the Xiaomi Redmi K50i display in sRGB colour profile using Spectral C6 Colorim-

eter and Calman Ultimate software. The K50i delivered 673.7 Nits of white luminance which is on the higher side when compared to IPS LCD panels in the mid-range segment. This falls in line with our expectations of higher transmittance from FFS LCD displays.

Calman recorded an average delta error of 2.2 in the sRGB colour mode, which is again quite good. The greyscale performance was also better than expected.

The red bias increases as we move towards 100 per cent white, which results in the sRGB mode being warmer than standard. The contrast ratio of 974 is again above average, but not particularly exciting. The colour volume of the display also turned out quite decent compared to conventional IPS LCD options.

TO SUM IT UP

So, overall, the FFS LCD displays retain the key wide viewing angle benefits of IPS LCD panels and further offer higher transmittance, higher responsiveness, and better power efficiency. In our testing, the FFS LCD display on the Xiaomi Redmi K50i fared better than the average IPS LCD panels we see in the affordable and mid-range segment. It also allows Xiaomi to keep the cost competitive while also delivering a 144Hz high refresh rate display. But at the same time, AMOLED panels are still better for multimedia content consumption. Siddharth Chauhan aptly summarized his experience with the FFS LCD display in his Xiaomi Redmi K50i review: "It also offers wider viewing angles than normal LCD panels and has high response times. All said and done, it's still not up to the standards of an AMOLED screen which is a bummer in my opinion. The display is decent enough to watch something on YouTube or even a movie on Netflix since it is HDR10 and Dolby Vision compliant. The refresh rate can go up to 144Hz which is great since it allows for increased smoothness but not many games can benefit from such a fast refresh rate yet."